STATISTICS

Statistics Cheat Sheet

The concepts interviewers ask, explained in one line each, plus A/B testing step by step.

Core concepts

Concept	One line
Mean vs median	Use median when data is skewed (salaries, house prices).
Standard deviation	Average distance of values from the mean.
Normal distribution	68% within 1 SD, 95% within 2, 99.7% within 3.
Central Limit Theorem	Sample means are roughly normal for large samples, whatever the original shape.
Null hypothesis	The default claim: there is no effect.
p-value	Chance of seeing a result this extreme if the null is true. Below 0.05 usually means significant.
Type I error	False positive. You say there is an effect when there is none.
Type II error	False negative. You miss a real effect.
Confidence interval	Range likely to contain the true value, e.g. 95% CI.
Statistical power	Chance of detecting an effect that exists. Aim for 80%.
Correlation	-1 to 1, strength of linear link. Does not prove causation.
Outliers (IQR)	Below Q1 - 1.5×IQR or above Q3 + 1.5×IQR.

Which test to use

Question	Test
Compare 2 group means	t-test
Compare 3+ group means	ANOVA
Two categorical variables related?	Chi-square
Compare 2 conversion rates	Two-proportion z-test
Relationship between 2 numbers	Correlation / regression

A/B testing in 7 steps

1. Define one primary metric (e.g. conversion rate).
2. Write the hypothesis. H0: no difference.
3. Decide minimum detectable effect, significance (0.05) and power (0.8).
4. Calculate sample size before starting.
5. Randomly split users. Run for full weeks.
6. Do not peek and stop early.
7. Test significance, check guardrail metrics, then decide.

Correlation vs causation: ice cream sales and drowning both rise in summer. Temperature drives both. Always ask what third factor could explain the link.

10 questions asked in data interviews

1. Explain p-value to a non-technical manager.
2. When would you use median over mean?
3. What is the Central Limit Theorem and why does it matter?
4. Type I vs Type II error with a business example.
5. How do you pick a sample size for an A/B test?
6. Your test is significant but the lift is 0.1%. Ship it?
7. What is Simpson's paradox?
8. How do you detect outliers?
9. Difference between a population and a sample.
10. What is multicollinearity?

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